

telephone cable, coaxial cable or other similar cable is used between the remote unit and user (Fig. 3).

Architectures

Selection of network architecture to be implemented is a very important criterion for providing telecom services to the consumers. A telecom service provider considers aspects such as network location, the cost of deploying the network, subscriber density, existing outside plant, the return on investment (ROI) and many more things.

Basically, two types of architectures are available, namely, active and passive. Active architectures are sometimes referred to as home-run fibre and/or active star Ethernet. On the other hand, passive architectures,

which include passive optical networks (PONs), are the current choices. Each has its own pros and cons, and the final selection depends on the provider's unique requirements.

Active optical networks (AONs)

Home-run fibre (point-to-point). As the name suggests, active optical networks employ active or powered devices. A home-run fibre architecture is one in which a dedicated fibre from an optical line terminal (OLT) located in the central office of the telecom service provider connects in a point-to-point fashion to an optical network terminal (ONT) residing at subscriber's premises. Both OLTs and ONTs are active, or powered, devices,

and each is equipped with an optical laser. Subscribers can be located as far away from the central office of the service provider as 80km, and each subscriber is provided a dedicated fibre that provides full bi-directional communication.

Home-run fibre is the most flexible architecture over the long term. However, it may be less attractive when the physical layer costs are considered. Since a dedicated fibre is deployed to each premise, home-run fibre requires installation of lot more fibres than other options, with each fibre running the entire distance between the subscriber and the central office. The fibre cost and size of the fibre bundle at the OLT can make this network expensive and inconvenient in many service areas.

Active star Ethernet (point-to-point). Active star Ethernet (ASE) architecture is a point-to-point architecture in which multiple premises share one feeder fibre through a remote node located between the central office and the served customer's premises (Fig. 4). Environmentally hardened optical Ethernet electronics—switches or broadband loop carriers—are installed at the remote node to provide fibre access aggregation. The remote node can be shared between four to a thousand homes via dedicated distribution links from the remote node.

Similar to the home-run fibre, subscribers can be located as far away from the remote node as 80km, and each subscriber is provided a dedicated fibre that provides full bidirectional bandwidth. ASE reduces the amount of fibre deployed, lowering costs through the sharing of fibre. ASE also offers the benefits of standard optical Ethernet technology, much simpler network topologies and supports a wide range of CPE solutions. And, most importantly, it provides broad flexibility for future growth.

Passive optical network (point-to-multipoint)

Passive optical network (PON) is essentially a cost-effective optical-fibre-based access system for providing multi-play services (voice, video, data, etc) to both

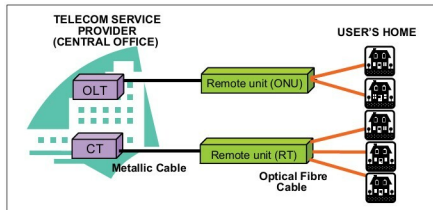


Fig. 3: FTTC configuration

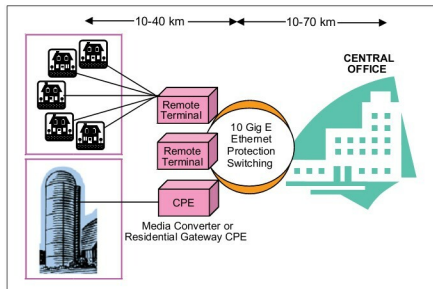


Fig. 4: Active star Ethernet architecture